

Contents

Week 2 work	3
Importing Required Libraries	3
Figure 1	3
Loading the Dataset	3
Display First Five Rows	3
Figure 2	3
Display Last Five Rows	4
Dataset Shape	4
Figure 3	4
Display Column Names	4
Figure 4	4
Data Types.....	4
Figure 5	5
Summary Statistics.....	5
Figure 6	5
Missing Values	6
Figure 7	6
Duplicate Records	6
Figure 8	6
Dataset Description.....	7
Feature Engineering	7
Family Size.....	7
Figure 9	7
Is Alone.....	7
Figure 10	8
Age Group	8
Figure 11	8
Fare Category.....	9
Figure 12	9
Feature Engineering Summary.....	9
Figure 13	9
Part 3: EDA Question 1: Passenger Survival Percentage	10
Figure 14	10
Question 2: Survival Rate by Gender	11

Figure 15	11
Question 3: Survival Rate by Passenger Class.....	11
Figure 16	12
Question 4: Does Age Affect Survival?.....	12
Figure 17	13
Figure 18	13
Question 5: Does Fare Influence Survival?	13
Figure 19	14
Question 6: Did Passengers Travelling Alone Survive More Often?	14
Figure 20	15
Question 7: Survival Rate by Embarkation Port.....	15
Figure 21	15
Question 8: Relationship Between Passenger Class and Fare	15
Figure 22	16
Question 9: Correlation Heatmap.....	16
Figure 23	17
Question 10: Family Size vs Survival	17
Figure 24	18
Part 4: Statistical Analysis	18
Figure 25	19
Part 5: Dashboard.....	19
Figure 26	19
Part 6: Business Insights	19
Overall Finding	20
Part 7: Business Recommendations	20

Week 2 work

Importing Required Libraries

The required Python libraries were imported to perform data manipulation, feature engineering, and data visualization. Pandas was used for handling the dataset, NumPy for numerical operations, Matplotlib for plotting graphs, and Seaborn for creating visually appealing statistical charts.

✓ Import Required Libraries

In this section, the required Python libraries are imported to perform data analysis, visualization, and feature engineering.

```
[1]
✓ 2s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Figure 1

Loading the Dataset

The cleaned Titanic dataset prepared during Week 1 was imported into Google Colab using the Pandas library. This dataset serves as the foundation for performing Exploratory Data Analysis and generating business insights.

Display First Five Rows

✓ Part 1 – Data Understanding

The purpose of this section is to understand the structure, content, and quality of the dataset before performing Exploratory Data Analysis (EDA).

✓ Display First Five Rows

The first five records of the dataset are displayed to understand the structure and available information.

[5]
✓ 0s
df.head()

	Passenger_ID	Survived	Passenger_Class	Name	Sex	Age	Siblings_Spouses	Parents_Children	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2: 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

Next steps: [Generate code with df](#) [New interactive sheet](#)

Figure 2

The head() function displays the first five records of the dataset, allowing us to understand the available features and verify that the dataset has been loaded correctly.

Display Last Five Rows

The `tail()` function displays the last five records of the dataset, helping verify the completeness and structure of the data.

Dataset Shape

The cleaned dataset contains **891 rows** and **11 columns**, indicating that all passenger records were retained while one unnecessary column (Cabin) was removed during data cleaning.

Dataset Shape

The shape of the dataset is checked to determine the total number of rows and columns.

```
[7]
✓ Os df.shape
... (891, 11)
```

Figure 3

Display Column Names

The dataset contains descriptive columns representing passenger information such as age, gender, ticket fare, passenger class, survival status, and embarkation port.

Column Names

The names of all columns are displayed to understand the available features used for analysis.

```
[8]
✓ Os df.columns
... Index(['Passenger_ID', 'Survived', 'Passenger_Class', 'Name', 'Sex', 'Age',
          'Siblings_Spouses', 'Parents_Children', 'Ticket', 'Fare', 'Embarked'],
          dtype='object')
```

Figure 4

Data Types

The `info()` function confirms the data types of each feature and provides an overview of the dataset structure. All columns have appropriate data types suitable for analysis.

Dataset Information

The information function is used to verify data types and ensure that each feature is stored in an appropriate format

```
[9] df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 11 columns):
#   Column             Non-Null Count  Dtype
---  -
0   Passenger_ID       891 non-null    int64
1   Survived           891 non-null    int64
2   Passenger_Class    891 non-null    int64
3   Name               891 non-null    object
4   Sex                891 non-null    object
5   Age                891 non-null    float64
6   Siblings_Spouses   891 non-null    int64
7   Parents_Children   891 non-null    int64
8   Ticket             891 non-null    object
9   Fare               891 non-null    float64
10  Embarked           891 non-null    object
dtypes: float64(2), int64(5), object(4)
memory usage: 76.7+ KB
```

Figure 5

Summary Statistics

Summary statistics provide important numerical information such as count, mean, standard deviation, minimum, maximum, and quartiles for numerical features. These values help in understanding the distribution of the data.

Summary Statistics

Descriptive statistics are generated to understand the numerical characteristics of the dataset.

```
[10] df.describe()

```

	Passenger_ID	Survived	Passenger_Class	Age	Siblings_Spouses	Parents_Children	Fare
count	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.361582	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	13.019697	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	22.000000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	35.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Figure 6

Missing Values

The dataset is checked for any remaining missing values after the cleaning process.

```
[11] df.isnull().sum()
✓ 0s
```

...	0
Passenger_ID	0
Survived	0
Passenger_Class	0
Name	0
Sex	0
Age	0
Siblings_Spouses	0
Parents_Children	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

Figure 7

The final missing value check confirms that all missing values were successfully handled during Week 1, and the dataset is now complete.

Duplicate Records

Figure 8

Duplicate records are checked to ensure data consistency.

```
[12] df.duplicated().sum()
✓ 0s
```

```
... np.int64(0)
```

No duplicate records were found in the cleaned dataset, confirming that each row represents a unique passenger.

Dataset Description

The Titanic dataset contains information about passengers who travelled on the Titanic, including their demographic details, ticket information, and survival status. The cleaned dataset consists of **891 rows and 11 columns** after preprocessing. It includes both numerical and categorical features such as age, gender, passenger class, fare, and embarkation port. This dataset is used to explore survival patterns and discover meaningful insights through Exploratory Data Analysis (EDA).

Feature Engineering

Feature engineering is the process of creating new useful features from existing data. In this project, four new features were created to make the Titanic dataset easier to analyze and to identify relationships between passenger characteristics and survival.

Family Size

A new **FamilySize** feature was created by adding the number of siblings/spouses, parents/children, and the passenger themselves.

Formula:

$$\text{FamilySize} = \text{Siblings_Spouses} + \text{Parents_Children} + 1$$

This feature helps analyze whether the size of a passenger's family group had any relationship with their chance of survival.



```
[10] ✓ 0s df[['Siblings_Spouses', 'Parents_Children', 'FamilySize']].head()
```

	Siblings_Spouses	Parents_Children	FamilySize
0	1	0	2
1	1	0	2
2	0	0	1
3	1	0	2
4	0	0	1

Figure 9

Is Alone

The **IsAlone** feature was created to identify whether a passenger was travelling alone or with family members. A passenger with a FamilySize of 1 was categorized as **Yes**, while passengers with a FamilySize greater than 1 were categorized as **No**.

This feature allows comparison of survival patterns between passengers travelling alone and those travelling with family.



[12] ✓ 0s df[["FamilySize", "IsAlone"]].head(10)

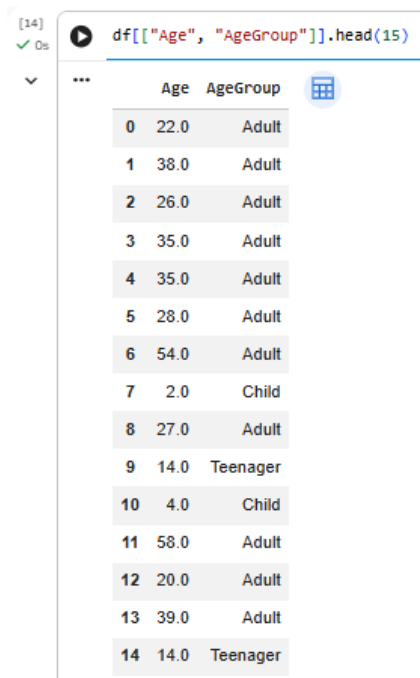
	FamilySize	IsAlone
0	2	No
1	2	No
2	1	Yes
3	2	No
4	1	Yes
5	1	Yes
6	1	Yes
7	5	No
8	3	No
9	2	No

Figure 10

Age Group

The **AgeGroup** feature was created by dividing passengers into four categories: **Child**, **Teenager**, **Adult**, and **Senior Citizen**.

This categorization makes it easier to compare survival patterns among different age groups rather than analyzing individual ages.



[14] ✓ 0s df[["Age", "AgeGroup"]].head(15)

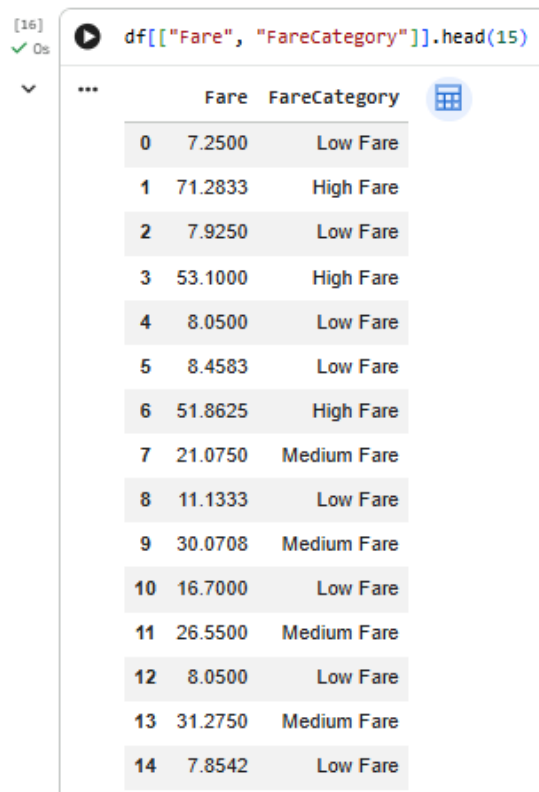
	Age	AgeGroup
0	22.0	Adult
1	38.0	Adult
2	26.0	Adult
3	35.0	Adult
4	35.0	Adult
5	28.0	Adult
6	54.0	Adult
7	2.0	Child
8	27.0	Adult
9	14.0	Teenager
10	4.0	Child
11	58.0	Adult
12	20.0	Adult
13	39.0	Adult
14	14.0	Teenager

Figure 11

Fare Category

The **FareCategory** feature was created by grouping passengers according to their ticket fare. Three categories were used: **Low Fare**, **Medium Fare**, and **High Fare**.

This feature helps investigate whether ticket price and passenger survival were related.



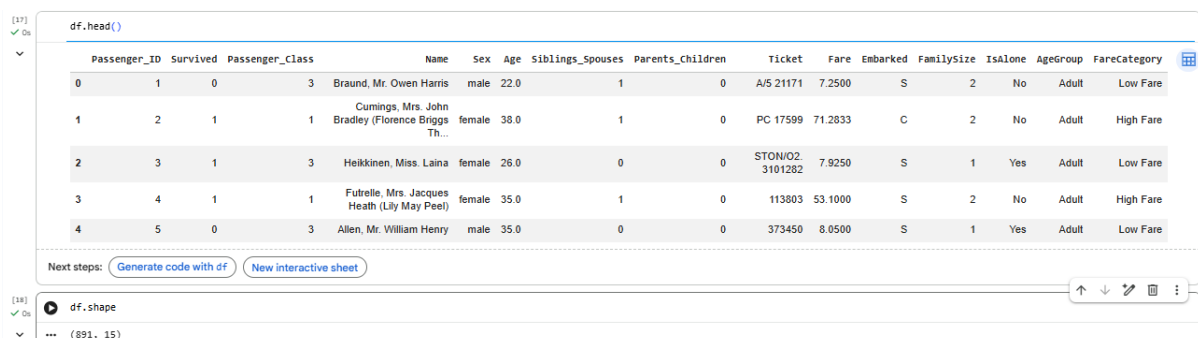
```
[16] ✓ Os df[["Fare", "FareCategory"]].head(15)
```

	Fare	FareCategory
0	7.2500	Low Fare
1	71.2833	High Fare
2	7.9250	Low Fare
3	53.1000	High Fare
4	8.0500	Low Fare
5	8.4583	Low Fare
6	51.8625	High Fare
7	21.0750	Medium Fare
8	11.1333	Low Fare
9	30.0708	Medium Fare
10	16.7000	Low Fare
11	26.5500	Medium Fare
12	8.0500	Low Fare
13	31.2750	Medium Fare
14	7.8542	Low Fare

Figure 12

Feature Engineering Summary

Four new features were successfully added to the dataset: **FamilySize**, **IsAlone**, **AgeGroup**, and **FareCategory**. These features provide additional information that will be used in the upcoming Exploratory Data Analysis to identify meaningful patterns and relationships in the Titanic dataset.



```
[17] ✓ Os df.head()
```

	Passenger_ID	Survived	Passenger_Class	Name	Sex	Age	Siblings_Spouses	Parents_Children	Ticket	Fare	Embarked	FamilySize	IsAlone	AgeGroup	FareCategory
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	2	No	Adult	Low Fare
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C	2	No	Adult	High Fare
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2 3101282	7.9250	S	1	Yes	Adult	Low Fare
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S	2	No	Adult	High Fare
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	1	Yes	Adult	Low Fare

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[18] ✓ Os df.shape
```

(891, 15)

Figure 13

Part 3: EDA

Question 1: Passenger Survival Percentage

The first analysis examined the overall survival distribution of Titanic passengers. The percentage of passengers who survived was calculated using the Survived column and presented using a pie chart.

The analysis showed that approximately **38.4% of passengers survived**, while **61.6% did not survive**. Therefore, the majority of passengers in the dataset did not survive the disaster.

```
[19] ✓ Os
survival_percentage = df["Survived"].value_counts(normalize=True) * 100

survival_percentage
```

	proportion
Survived	
0	61.616162
1	38.383838

dtype: float64

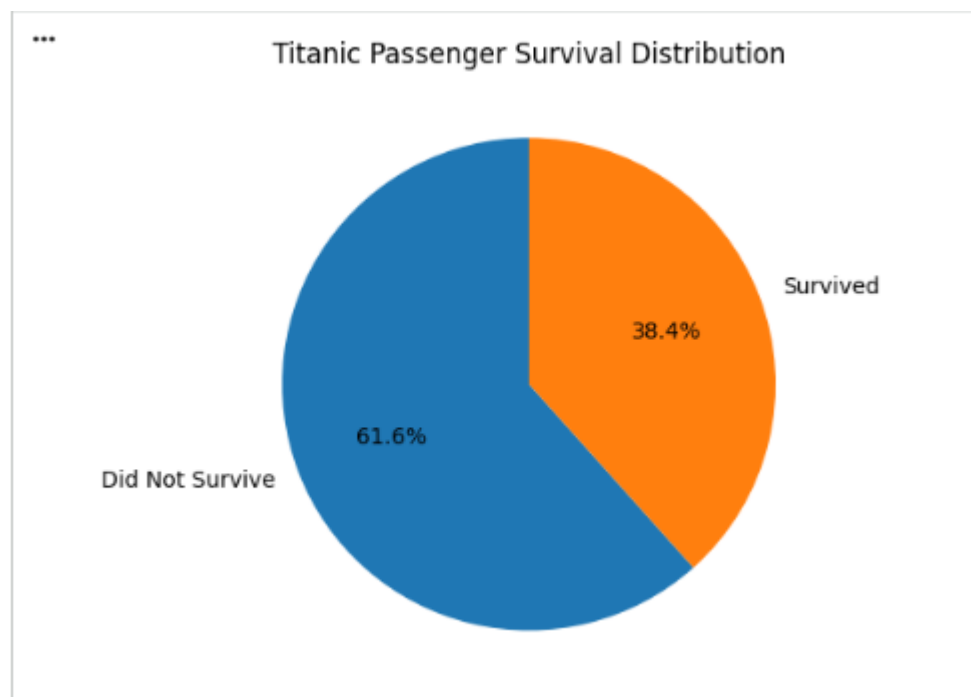


Figure 14

Question 2: Survival Rate by Gender

The survival rate was analyzed separately for male and female passengers to determine whether gender was associated with survival. A bar chart was used to compare the survival rates of both groups.

The analysis showed that female passengers had a significantly higher survival rate than male passengers. Approximately **74.2% of female passengers survived**, compared with approximately **18.9% of male passengers**. This indicates a strong difference in survival outcomes between the two genders.

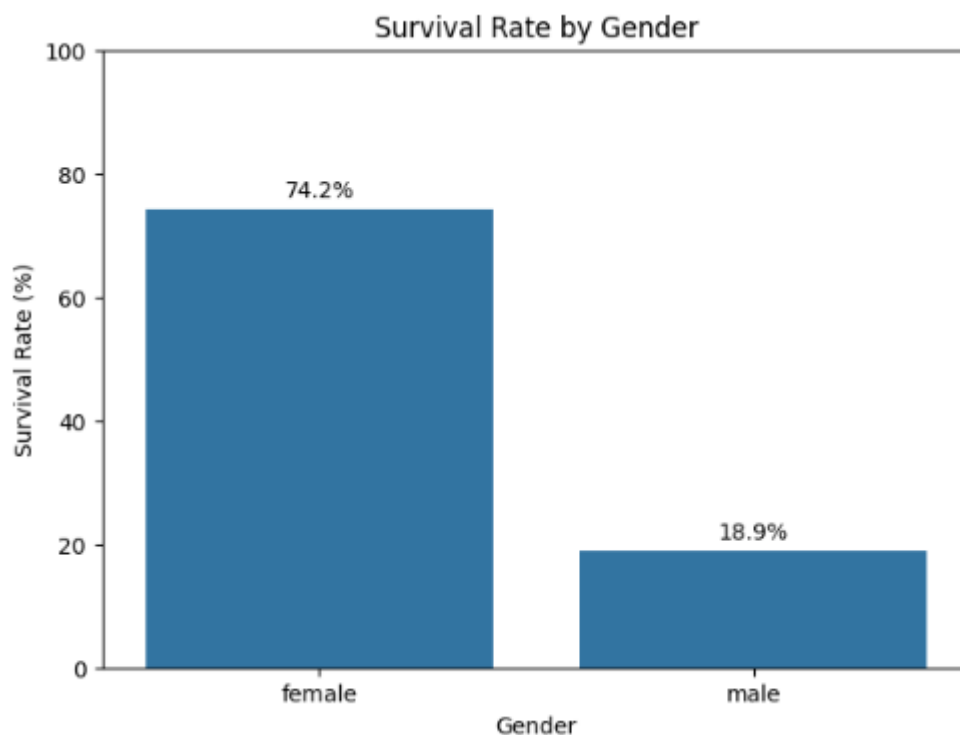


Figure 15

Question 3: Survival Rate by Passenger Class

The survival rates of passengers were compared across the three passenger classes using a bar chart. This analysis was performed to determine whether passenger class had a relationship with survival.

The results showed that **first-class passengers had the highest survival rate**, while third-class passengers had the lowest survival rate. This indicates that passengers belonging to higher classes generally had better survival outcomes than those in lower classes.

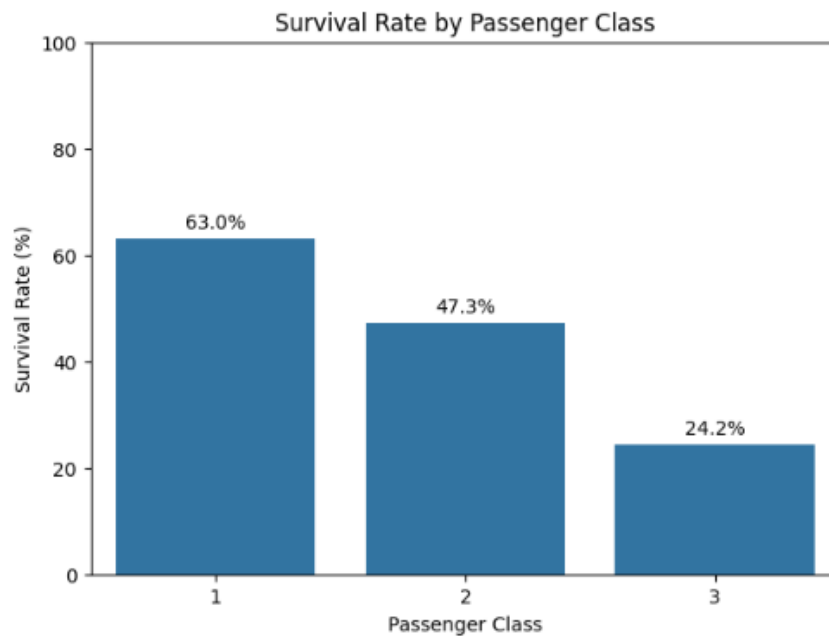


Figure 16

Question 4: Does Age Affect Survival?

The relationship between passenger age and survival was analyzed using the age distribution and age-group survival rates. The average age of passengers who survived was **28.29 years**, while the average age of passengers who did not survive was **30.03 years**. This difference is relatively small.

A stronger pattern was observed when passengers were divided into age groups. **Children had the highest survival rate at 57.97%**, followed by teenagers at **41.05%** and adults at **36.52%**. Senior citizens had the lowest survival rate at **26.92%**. These results suggest that age group had an influence on survival, with younger passengers generally having higher survival rates.

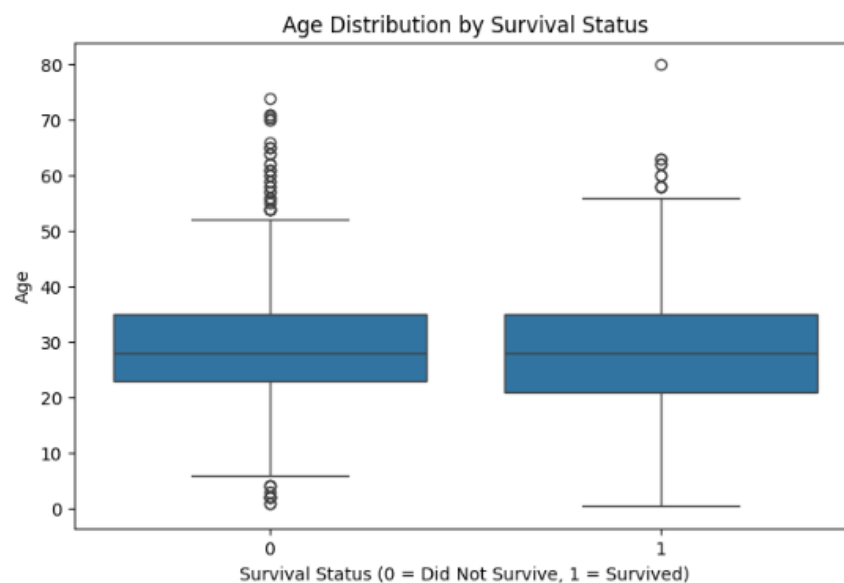
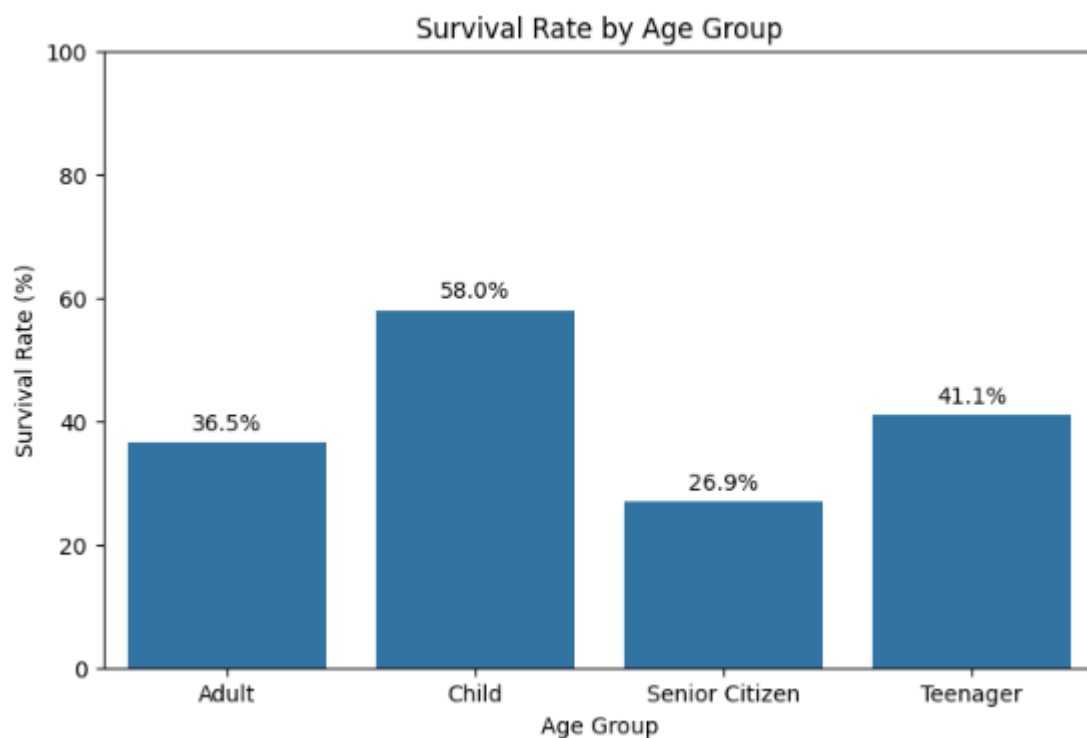


Figure 17

Figure 18



Question 5: Does Fare Influence Survival?

The relationship between ticket fare and passenger survival was analyzed using average fare and fare categories. The average fare paid by survivors was **48.40**, compared with **22.12** for passengers who did not survive.

When passengers were divided into fare categories, the **High Fare** group had the highest survival rate at **68.13%**. The **Medium Fare** group had a survival rate of **41.67%**, while the **Low Fare** group had the lowest survival rate at **27.77%**. These results indicate a positive association between fare level and survival. However, this relationship may also be influenced by passenger class because higher fares were generally associated with higher passenger classes.

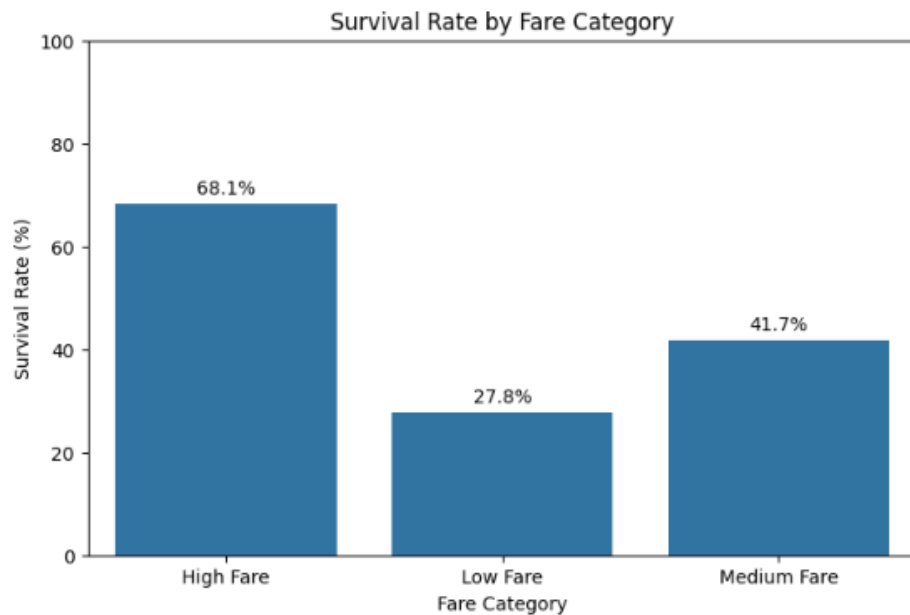


Figure 19

Question 6: Did Passengers Travelling Alone Survive More Often?

The relationship between travelling status and survival was analyzed using the `IsAlone` feature created during feature engineering. Passengers were divided into two groups: those travelling alone and those travelling with family members.

The results showed that passengers travelling with family had a higher survival rate of **50.56%**, compared with **30.35%** for passengers travelling alone. Therefore, passengers travelling with family survived more often in this dataset. This suggests that travelling with family was associated with better survival outcomes.

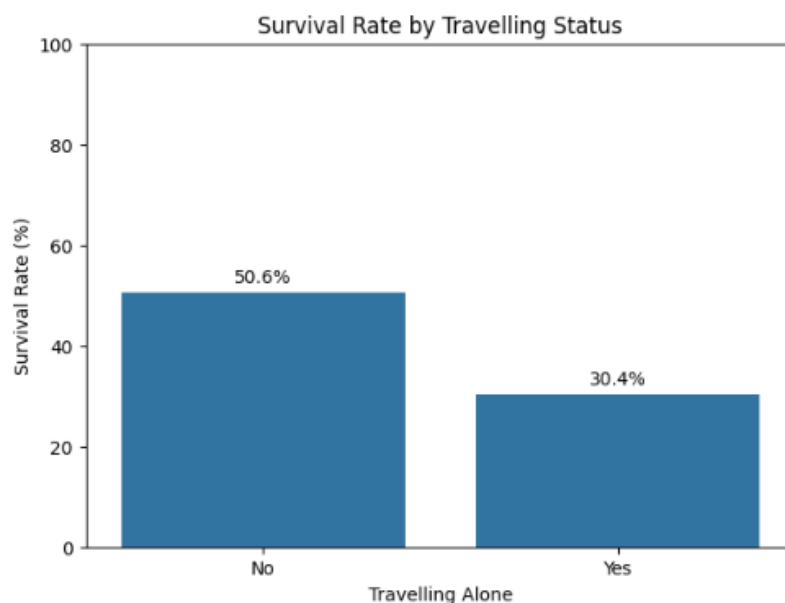


Figure 20

Question 7: Survival Rate by Embarkation Port

The survival rates of passengers were compared according to their embarkation ports. The analysis showed that passengers who embarked from **Cherbourg (C)** had the highest survival rate of **55.36%**. Passengers from **Queenstown (Q)** had a survival rate of **38.96%**, while passengers from **Southampton (S)** had the lowest survival rate of **33.90%**.

These results show that survival rates differed across embarkation ports. However, the embarkation port itself may not directly explain survival differences because other factors, such as passenger class and fare, could also influence the results.

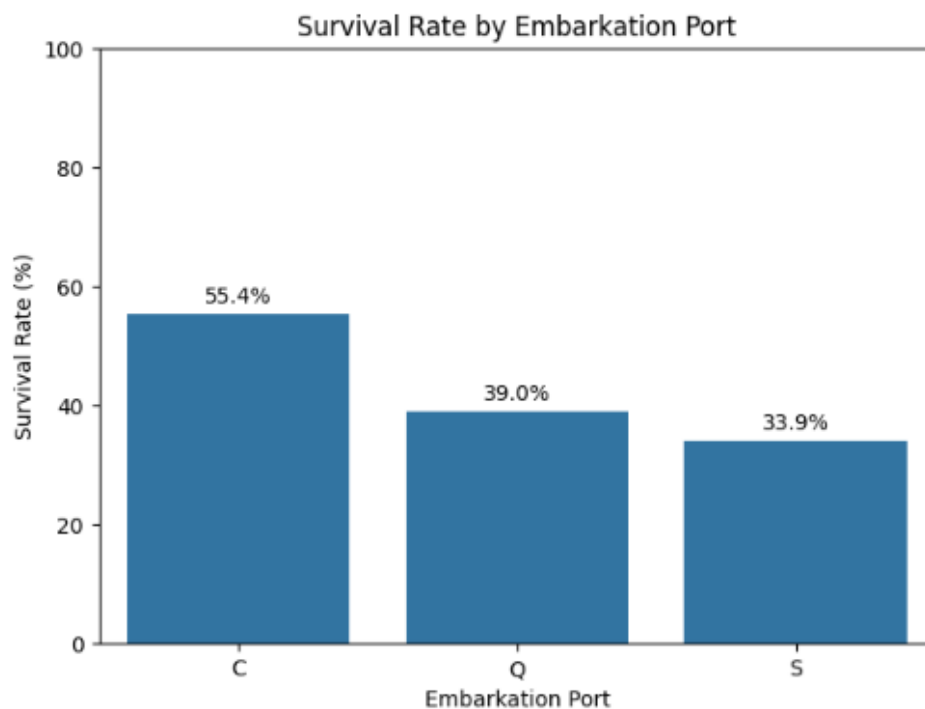


Figure 21

Question 8: Relationship Between Passenger Class and Fare

The relationship between passenger class and ticket fare was analyzed using a box plot and average fare calculations. The results showed a substantial difference in fare levels across the three passenger classes.

First-class passengers paid the highest average fare of **84.15**, followed by second-class passengers with an average fare of **20.66**. Third-class passengers paid the lowest average fare of **13.68**. The visualization also shows that fare values were generally higher for first-class passengers.

This finding is important because it helps explain the positive relationship observed earlier between fare and survival. Higher fares were strongly associated with higher passenger classes, which also had higher survival rates.

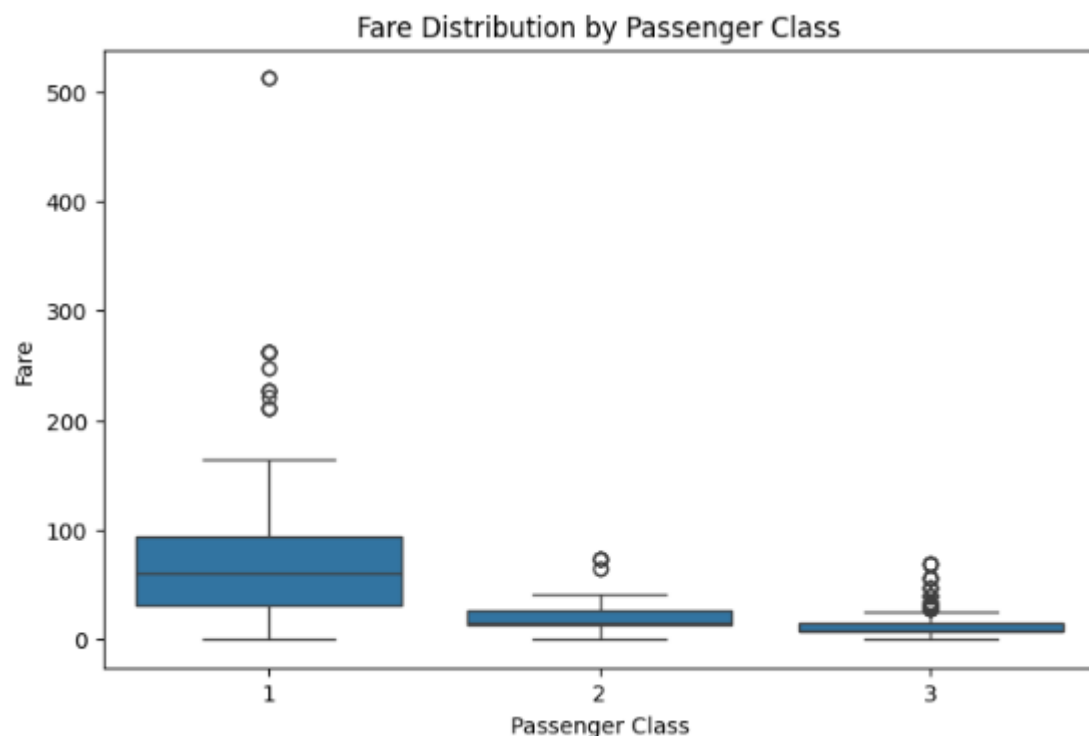


Figure 22

Question 9: Correlation Heatmap

A correlation heatmap was created to examine the relationships between numerical variables in the Titanic dataset. The strongest positive correlation between different variables was observed between **Siblings_Spouses and FamilySize (0.89)**, followed by **Parents_Children and FamilySize (0.78)**. These strong relationships are expected because FamilySize was created using Siblings_Spouses and Parents_Children.

The strongest negative correlation was observed between **Passenger_Class and Fare (-0.55)**. Since a lower Passenger_Class number represents a higher passenger class, this indicates that higher-class passengers generally paid higher fares. Passenger_Class also showed a negative correlation with Survival (-0.34), while Fare showed a positive correlation with Survival (0.26). These findings support the earlier analysis showing that passenger class and fare were associated with survival outcomes.

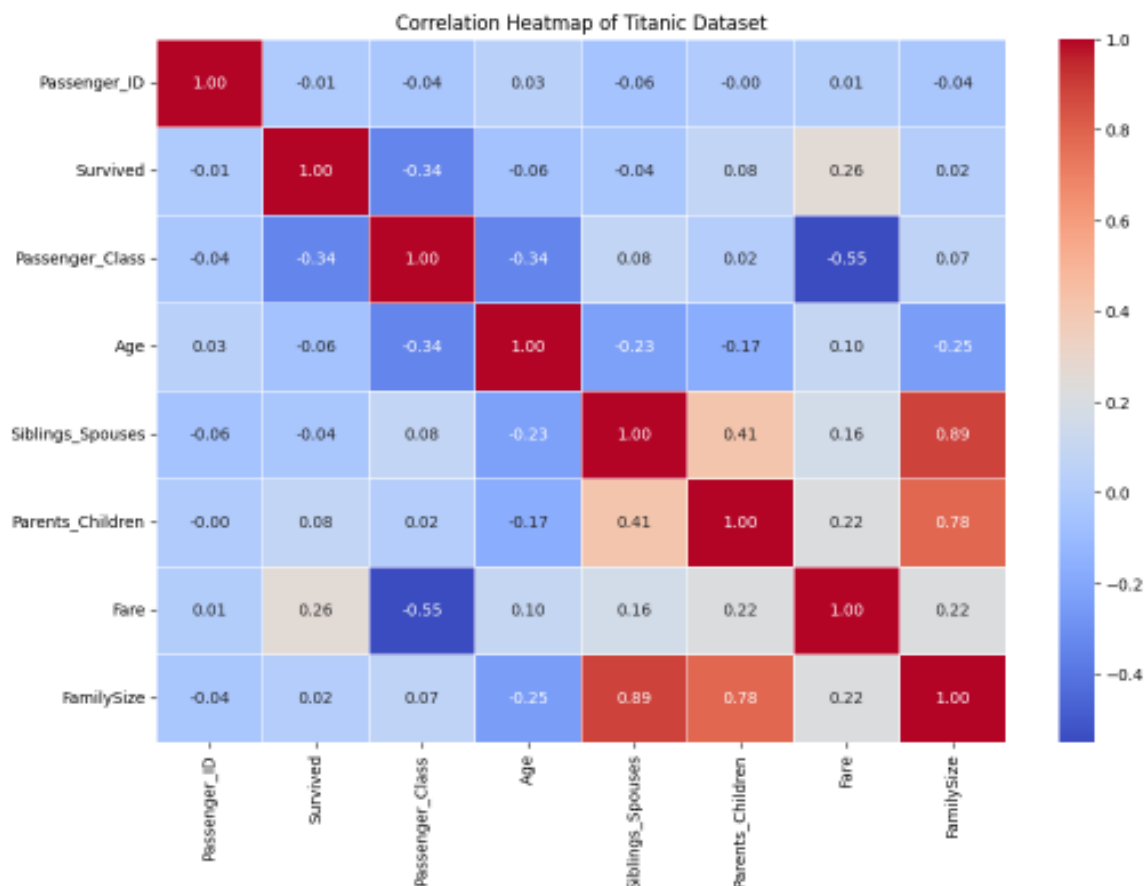


Figure 23

Question 10: Family Size vs Survival

The relationship between family size and passenger survival was analyzed using the FamilySize feature created during feature engineering. The analysis showed that passengers travelling in small to moderate family groups generally had higher survival rates.

Passengers with a family size of **4 had the highest survival rate of 72.41%**, followed by family sizes of 3 and 2, with survival rates of **57.84%** and **55.28%**, respectively. Passengers travelling alone had a survival rate of **30.35%**. Larger family groups, particularly sizes 5 and 6, had lower survival rates.

However, family sizes of 8 and 11 had very few passengers and a survival rate of 0%, so these results should be interpreted cautiously. Overall, the analysis suggests that **travelling with a small or moderate-sized family group was associated with better survival outcomes than travelling alone or in very large groups.**

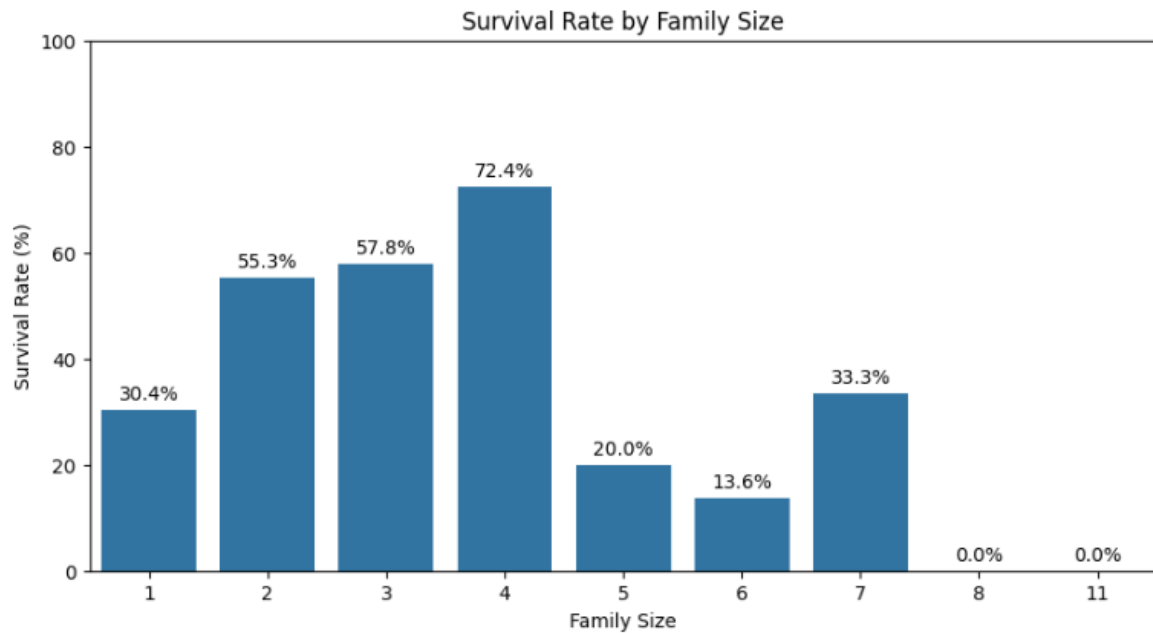


Figure 24

Part 4: Statistical Analysis

Descriptive statistics were calculated for three numerical variables: **Age**, **Fare**, and **FamilySize**. The mean, median, mode, and standard deviation were used to summarize the central tendency and variation of these variables.

Variable	Mean	Median	Mode	Standard Deviation
Age	29.36	28.00	28.00	13.02
Fare	32.20	14.45	8.05	49.69
FamilySize	1.90	1.00	1.00	1.61

Age: The average passenger age was 29.36 years, with a median and mode of 28 years. The standard deviation of 13.02 indicates considerable variation in passenger ages.

Fare: The average fare was 32.20, while the median was considerably lower at 14.45. This difference suggests that the fare distribution was right-skewed because some passengers paid substantially higher fares. The standard deviation of 49.69 also indicates considerable variation in ticket fares.

FamilySize: The average family size was 1.90, while the median and mode were both 1. This indicates that travelling alone was common in the dataset. The standard deviation of 1.61 shows some variation in family size.

Age
Mean: 29.36158249158249
Median: 28.0
Mode: 28.0
Standard Deviation: 13.019696550973194

Fare
Mean: 32.204207968574636
Median: 14.4542
Mode: 8.05
Standard Deviation: 49.693428597180905

FamilySize
Mean: 1.904601571268238
Median: 1.0
Mode: 1
Standard Deviation: 1.6134585413550788

Figure 25

Part 5: Dashboard

A dashboard was created to summarize the major findings from the exploratory data analysis. It includes survival distribution, gender distribution, passenger class distribution, age distribution, fare distribution, and a correlation heatmap.

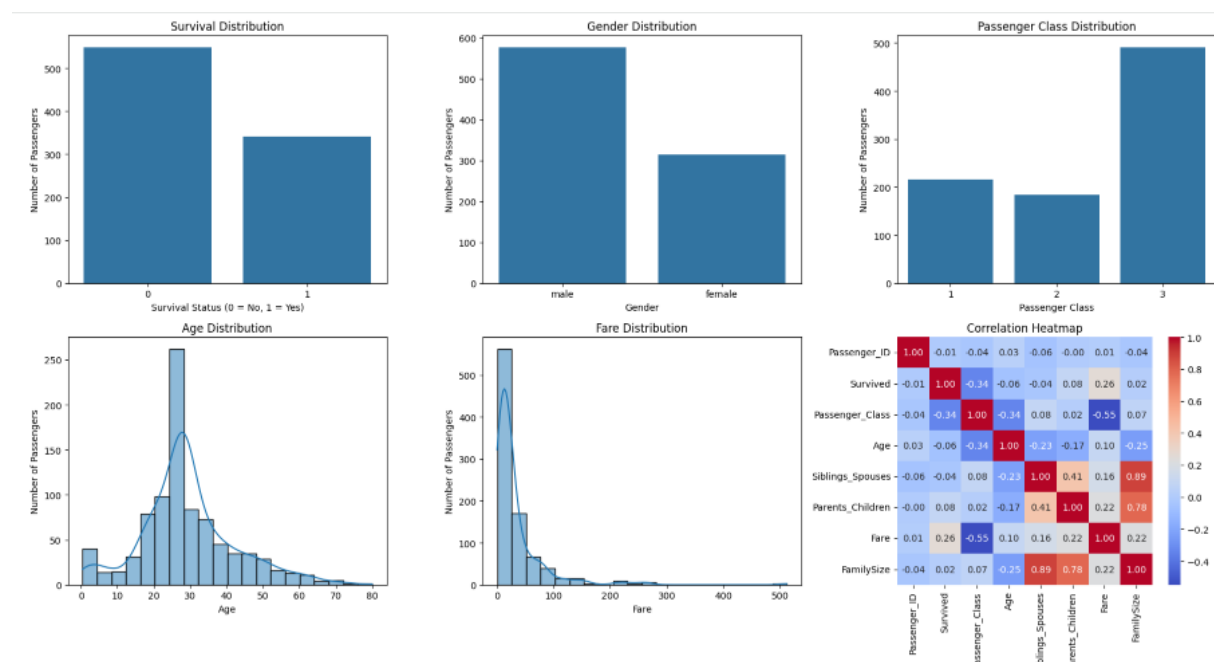


Figure 26

Part 6: Business Insights

1. Overall, **38.38%** of passengers survived.
2. Female passengers had a much higher survival rate (**74.20%**) than males (**18.89%**).

3. First-class passengers had the highest survival rate (**62.96%**).
4. Children had the highest survival rate among age groups (**57.97%**).
5. High-fare passengers had a higher survival rate (**68.13%**) than low-fare passengers (**27.77%**).
6. Passengers travelling with family had a higher survival rate (**50.56%**) than those travelling alone (**30.35%**).
7. Cherbourg had the highest survival rate among embarkation ports (**55.36%**).
8. First-class passengers paid considerably higher average fares (**84.15**) than second and third class.
9. Small to moderate family groups generally showed better survival outcomes than very large groups.
10. Passenger class and fare showed noticeable relationships with survival, supporting the patterns found in the earlier analyses.

Overall Finding

The analysis indicates that **gender, passenger class, fare, age group, and travelling status** were important factors associated with survival outcomes in the Titanic dataset.

Part 7: Business Recommendations

Based on the findings from the exploratory analysis, the following recommendations are proposed:

1. **Improve emergency support for vulnerable passengers**, particularly children and elderly passengers who may require additional assistance during emergencies.
2. **Strengthen safety procedures across all passenger classes** to ensure that emergency resources and evacuation opportunities are accessible to passengers regardless of their class.
3. **Develop organized family evacuation procedures** to help passengers travelling with family remain together during emergency situations.
4. **Provide additional assistance to solo travellers**, as passengers travelling alone had a lower survival rate than passengers travelling with family.
5. **Use passenger characteristics in emergency planning**, including age, passenger class, family size, and travelling status, to identify groups that may require additional support.